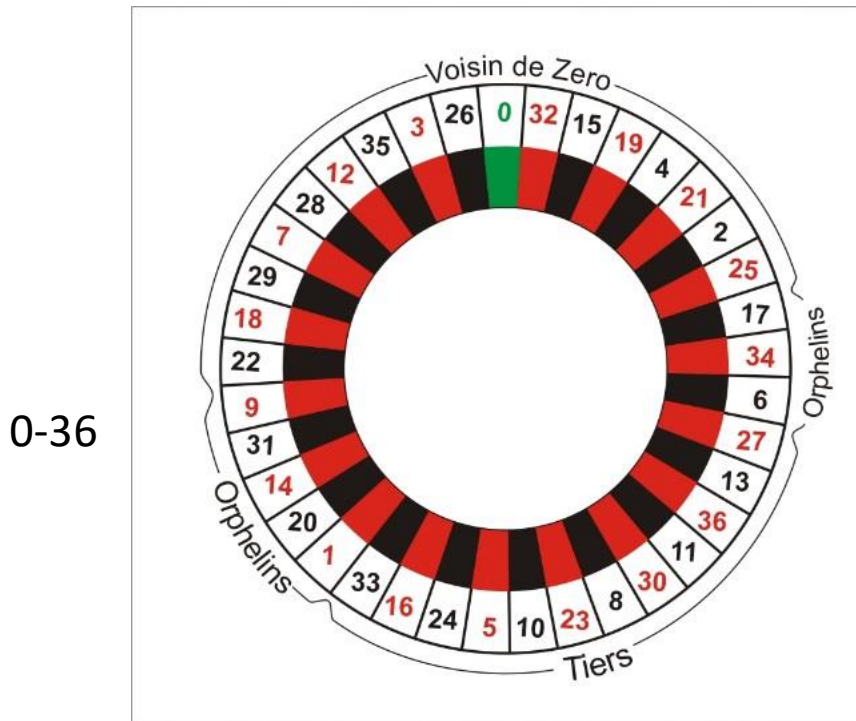


The Normal distribution

IQ and the bell curve

Simple distributions

Usually, when we say random we mean the Uniformly random:



0-1

0-59



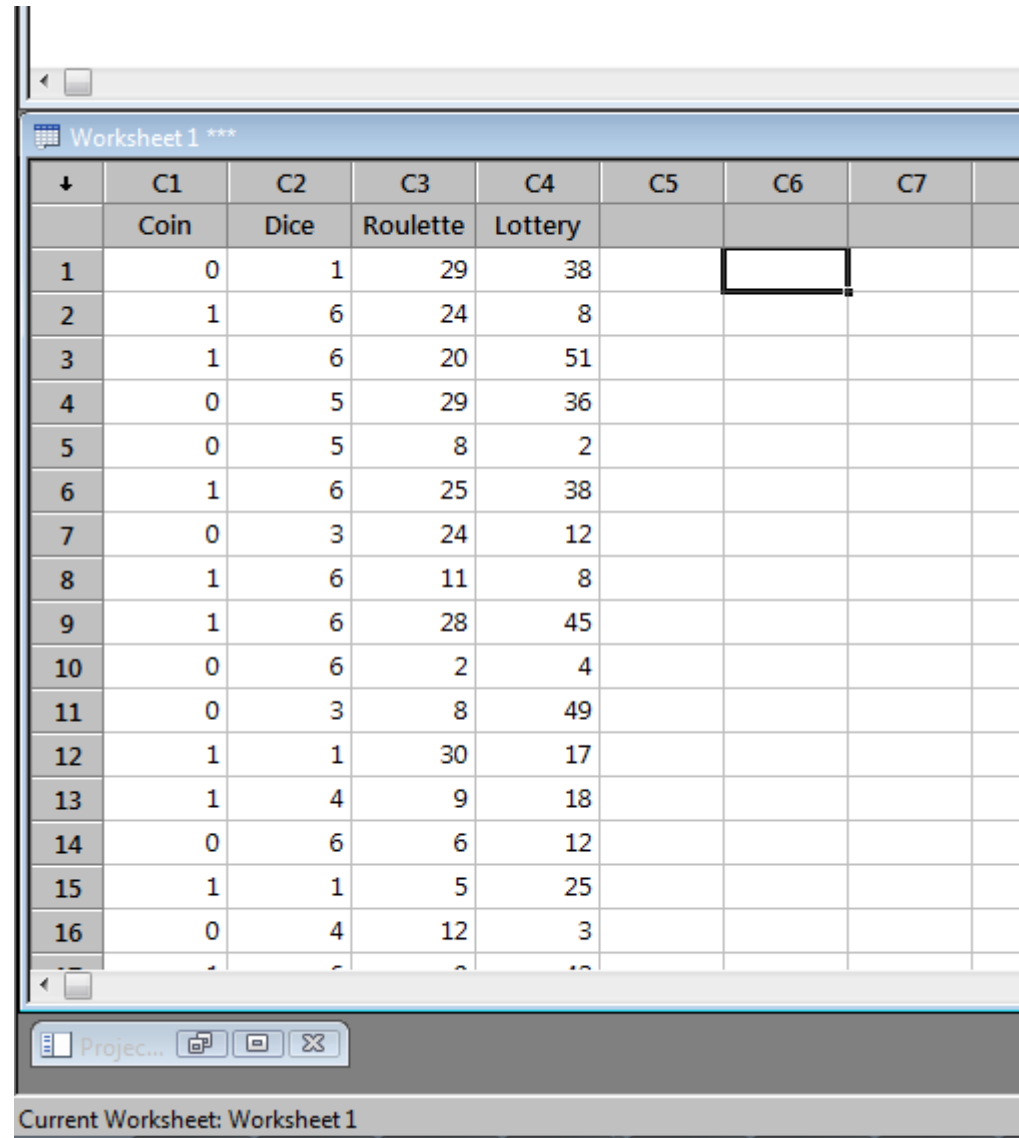
1-6

Uniformly random

100,000 rows of data

Generated using Minitab:

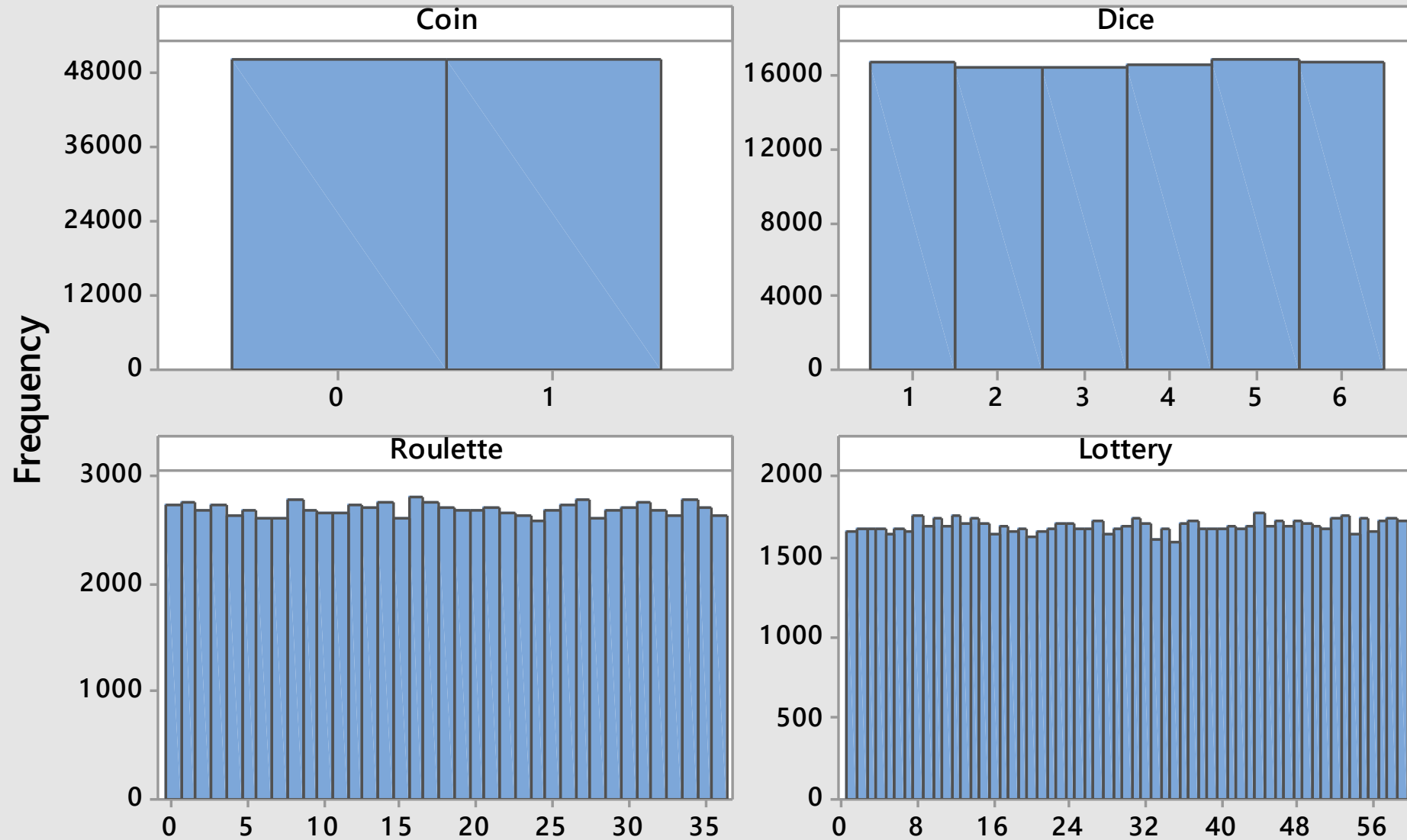
Calc>random data>Integer



The screenshot shows a Minitab worksheet titled "Worksheet 1 ***". The worksheet contains 16 rows of data. The first row is a header with labels: "Coin", "Dice", "Roulette", "Lottery", and empty cells for "C5", "C6", and "C7". The subsequent 15 rows contain random integer values. The status bar at the bottom indicates "Current Worksheet: Worksheet 1".

	C1	C2	C3	C4	C5	C6	C7
	Coin	Dice	Roulette	Lottery			
1	0	1	29	38			
2	1	6	24	8			
3	1	6	20	51			
4	0	5	29	36			
5	0	5	8	2			
6	1	6	25	38			
7	0	3	24	12			
8	1	6	11	8			
9	1	6	28	45			
10	0	6	2	4			
11	0	3	8	49			
12	1	1	30	17			
13	1	4	9	18			
14	0	6	6	12			
15	1	1	5	25			
16	0	4	12	3			

Uniform probability densities



C6	C7	C8	C9	C10	C11
Coin 1	Coin 2	Coin 3	double coin	triple coin	
1	1	0	2		
1	0	1	1		
0	1	0	1		
0	0	1	0		
1	0	1	1		
0	1	0	1		
1	1	0	2		
1	0	1	1		
0	1	1	1		
1	0	1	1		
1	1	0	2		
1	1	1	2		
1	1	0	2		
1	0	0	1		
1	1	0	2		
0	1	1	1		

Row Statistics

Statistic

- ☒ Sum
- ☐ Mean
- ☐ Standard deviation
- ☐ Minimum
- ☐ Maximum
- ☐ Range
- ☐ Median
- ☐ Sum of squares
- ☐ N total
- ☐ N nonmissing
- ☐ N missing

Input variables:

C11-C110

Store result in: 'Coin x 100'

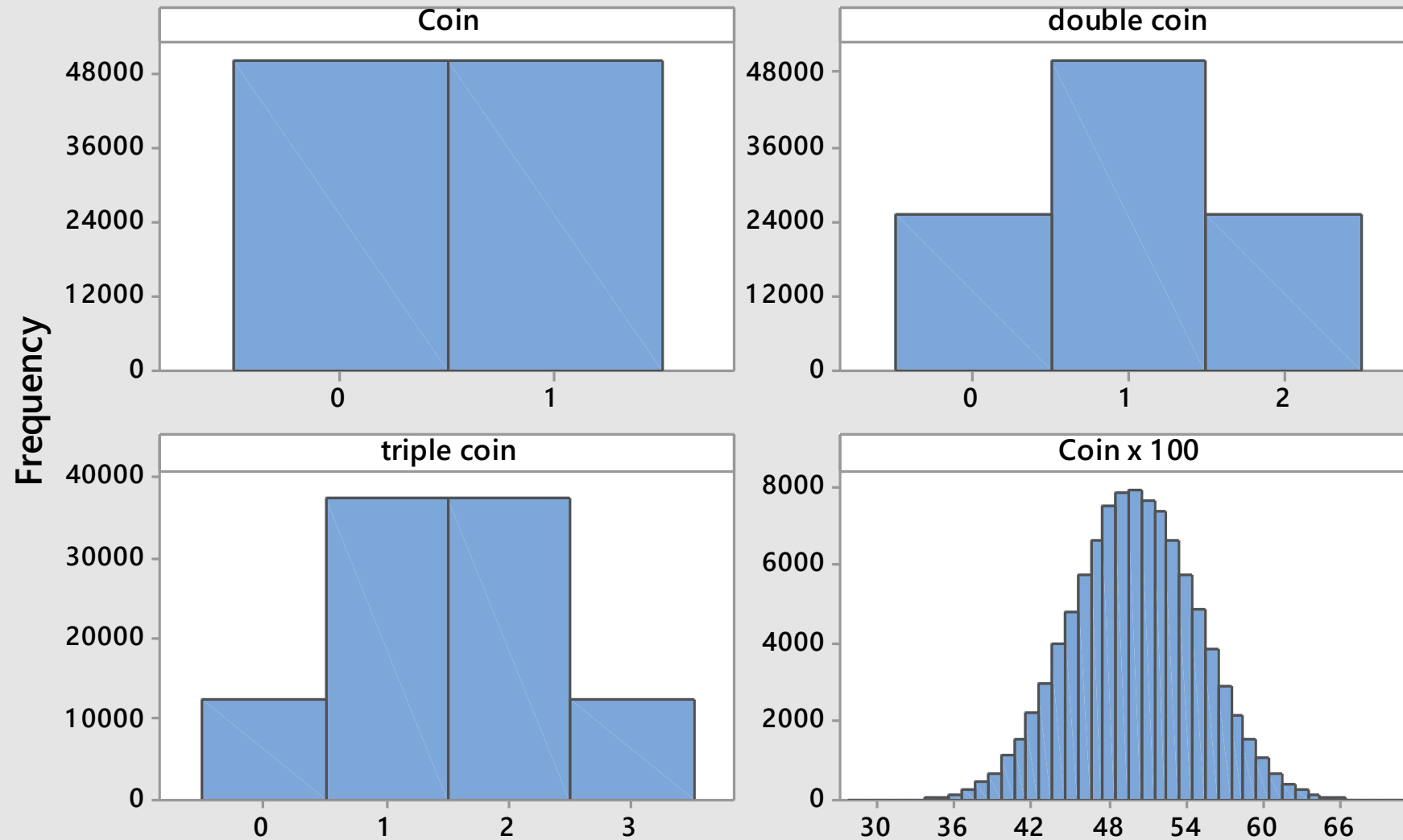
Select

Help

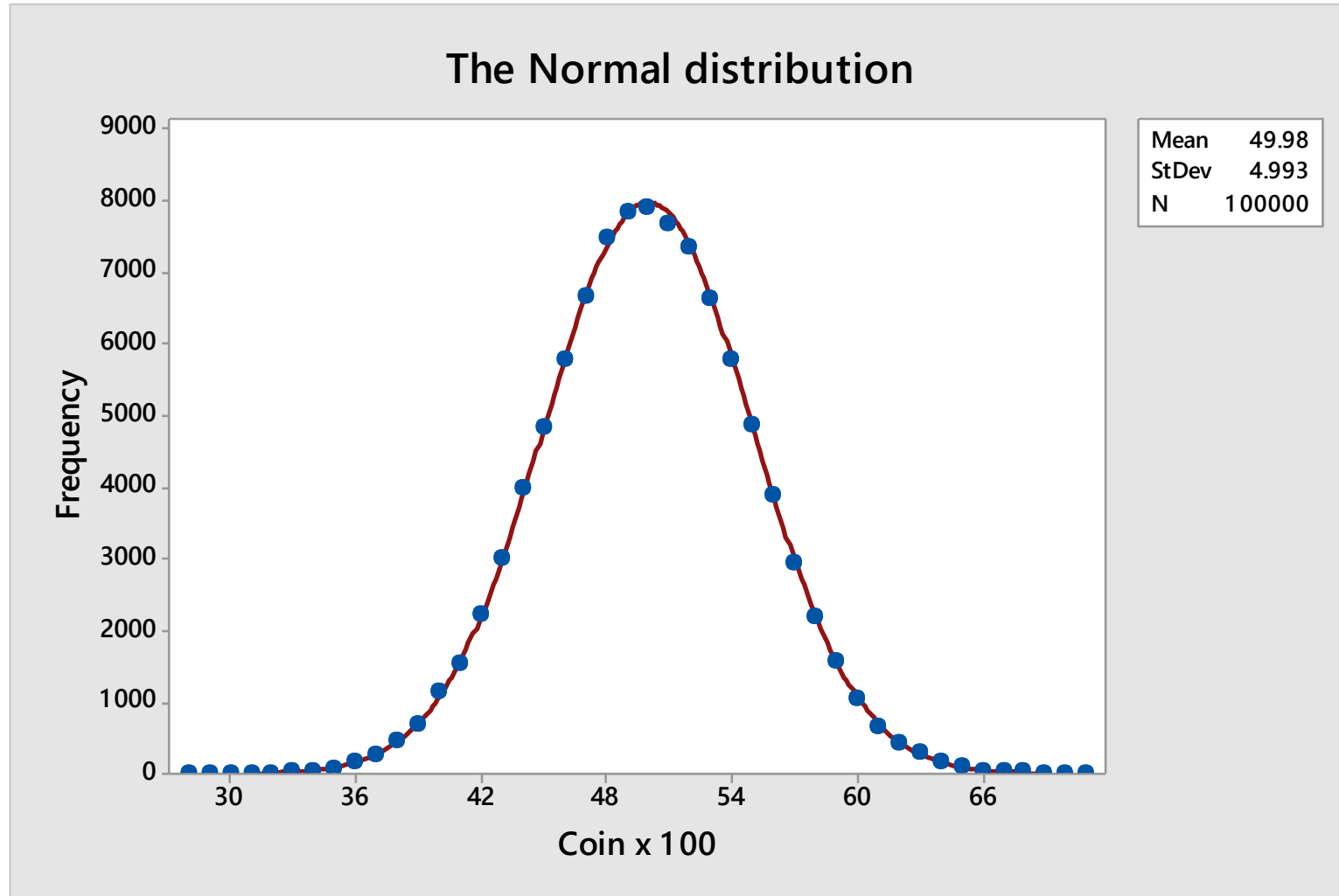
OK

Cancel

frequency distributions when summed

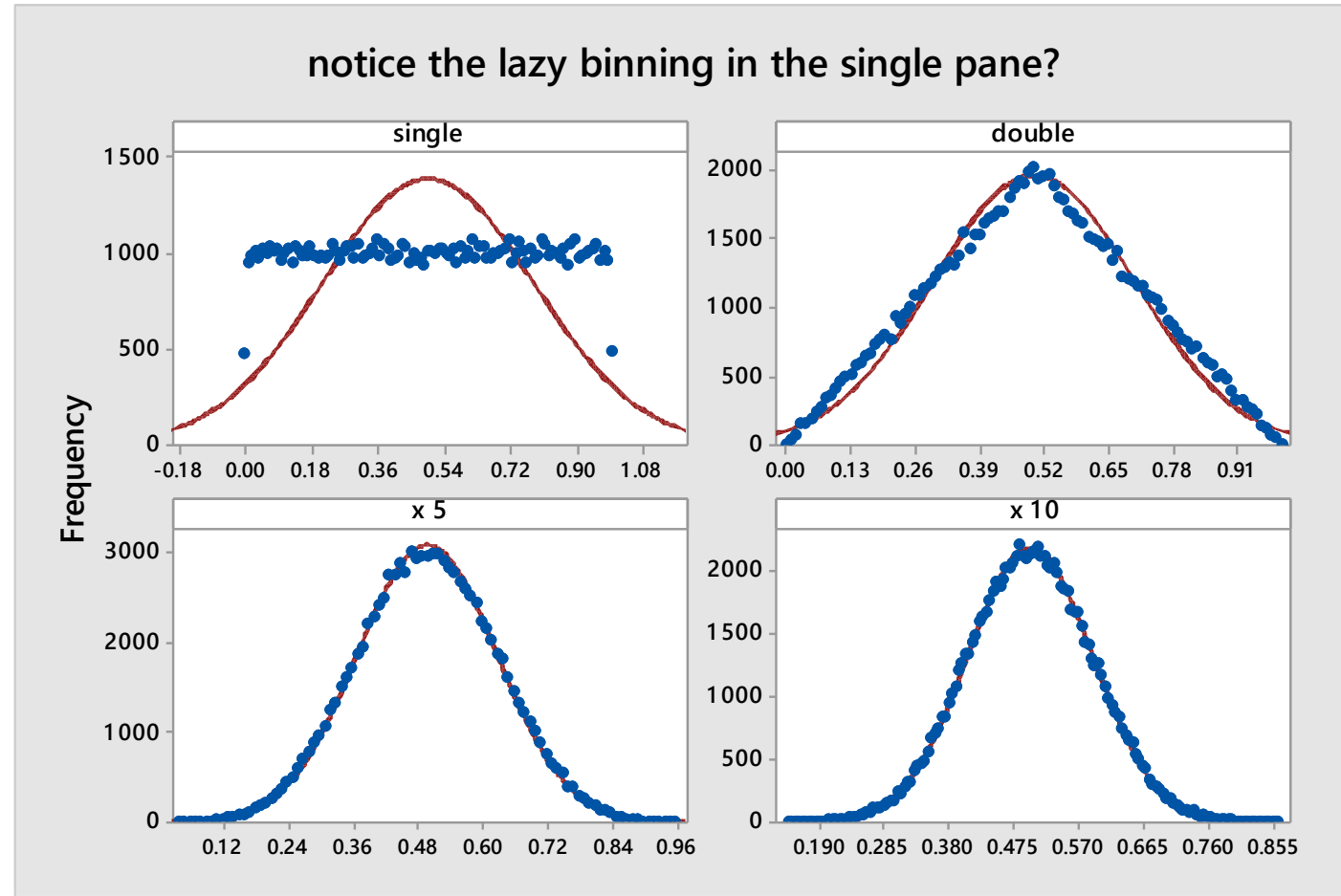


The Normal Distribution



What is Normal?

Any sum (or **average**) of **multiple** identically distributed distributions



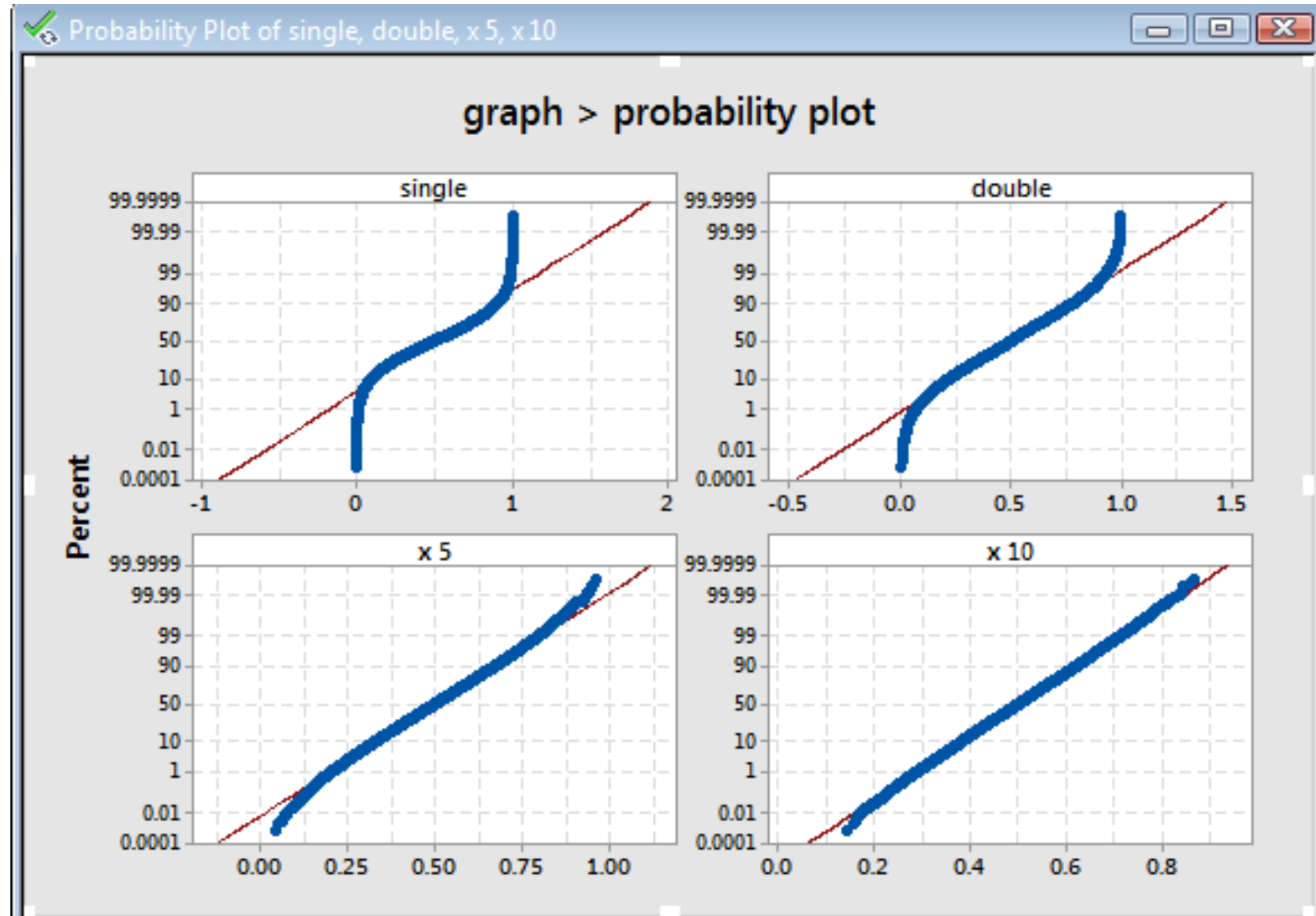
What variables are normally distributed?

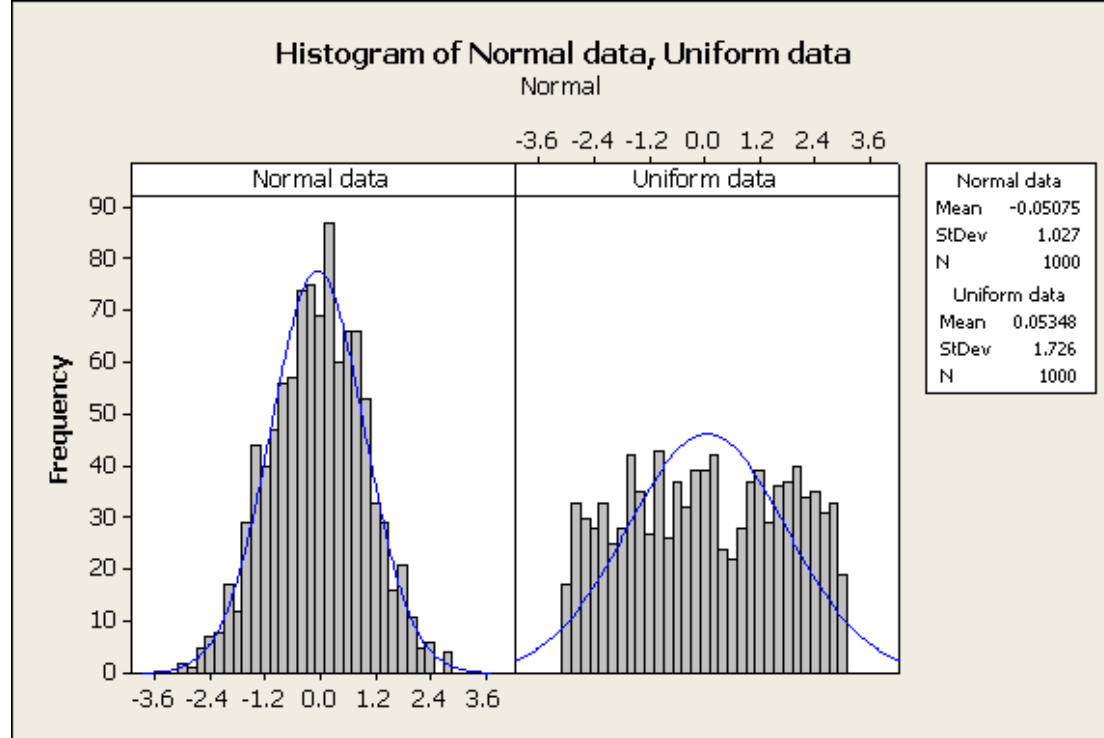
Many *averaged* qualities: **IQ** Rainfall, molecular velocities, football crowd attendance, test scores, women's height etc.

Some are definitely **not** normal. Key indicators are asymmetry and side-limited ranges: Wages, people's height, etc.



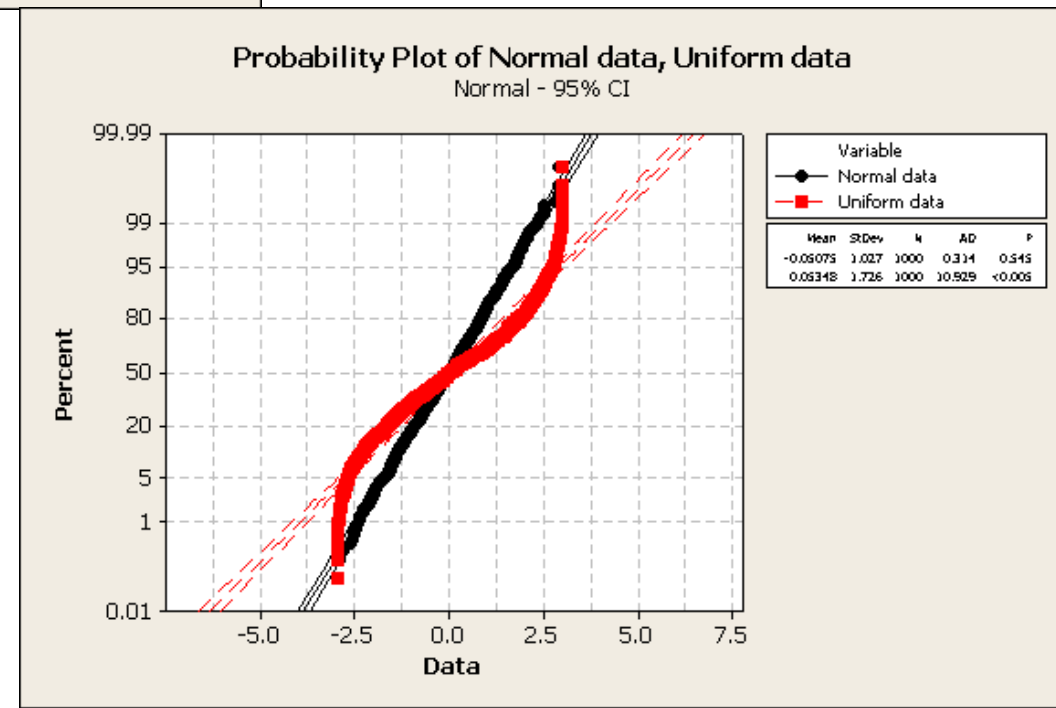
Checking for Normality



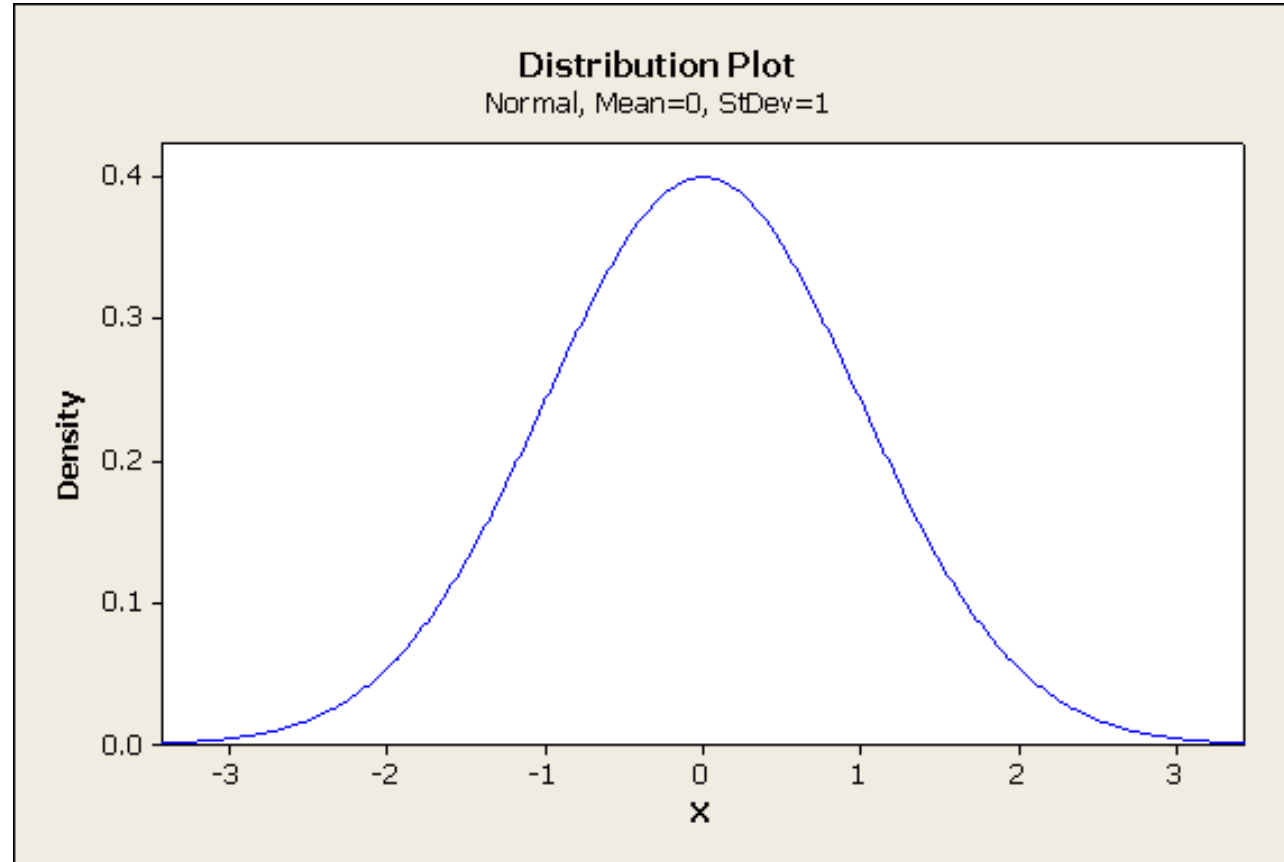


Compare n=1000
normally and
uniformly distributed
datasets

Difference clearer on
probability plot



The Normal distribution



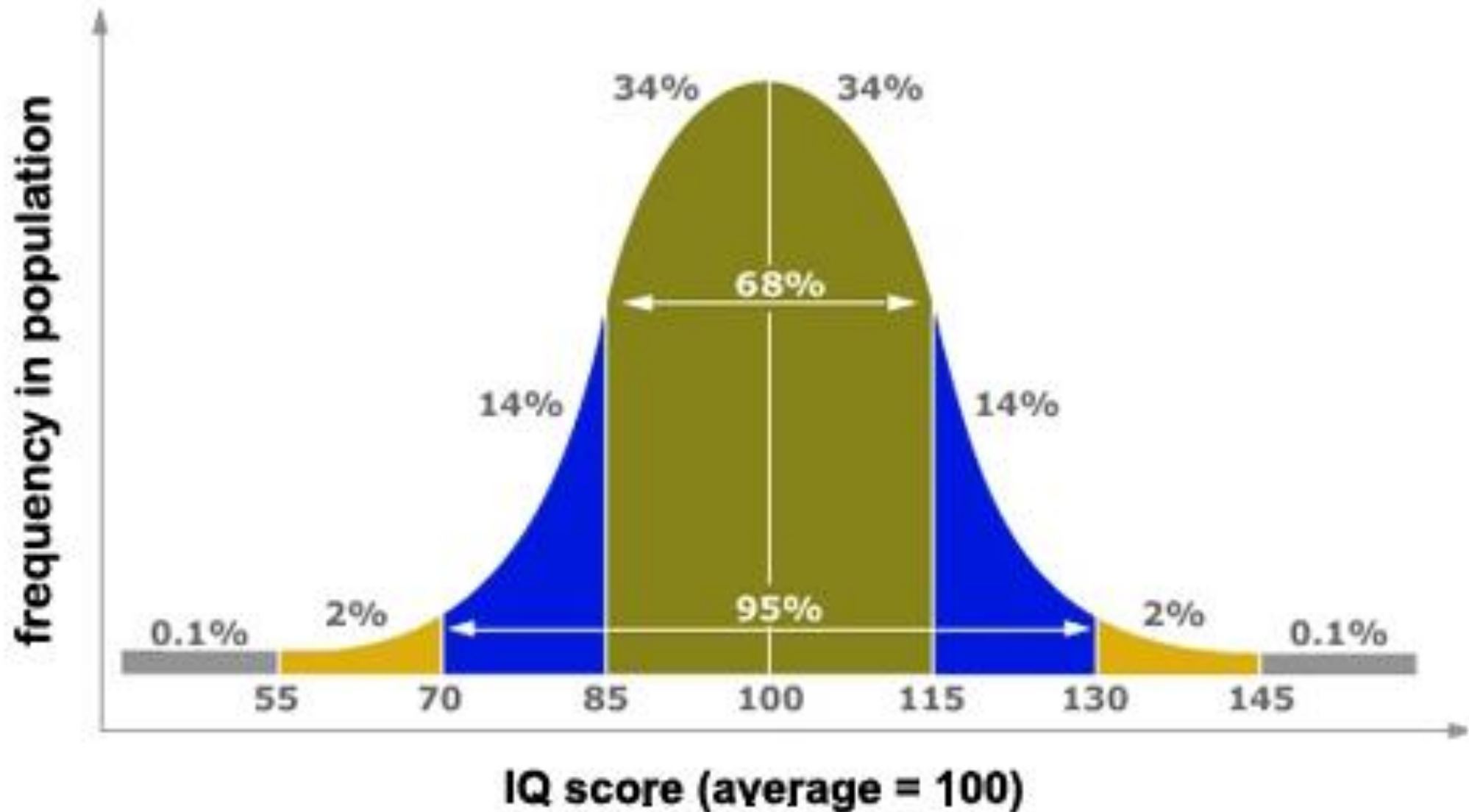
Symmetrical, central maximum, **vanishing tails**

What equation governs the line?

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2}$$

with $\mu = 0$, $\sigma = 1$ we have: $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

Often called the Bell curve



The book - 1994

The Bell curve – Herrnstein & Murray

Premise:

- IQ is a better predictor of job success, criminal activity etc. than social background
- IQ is normally distributed because it is the sum of many small variations in genes and experience

Controversy: suggests different racial groups have different IQ scores

Is IQ genetic?

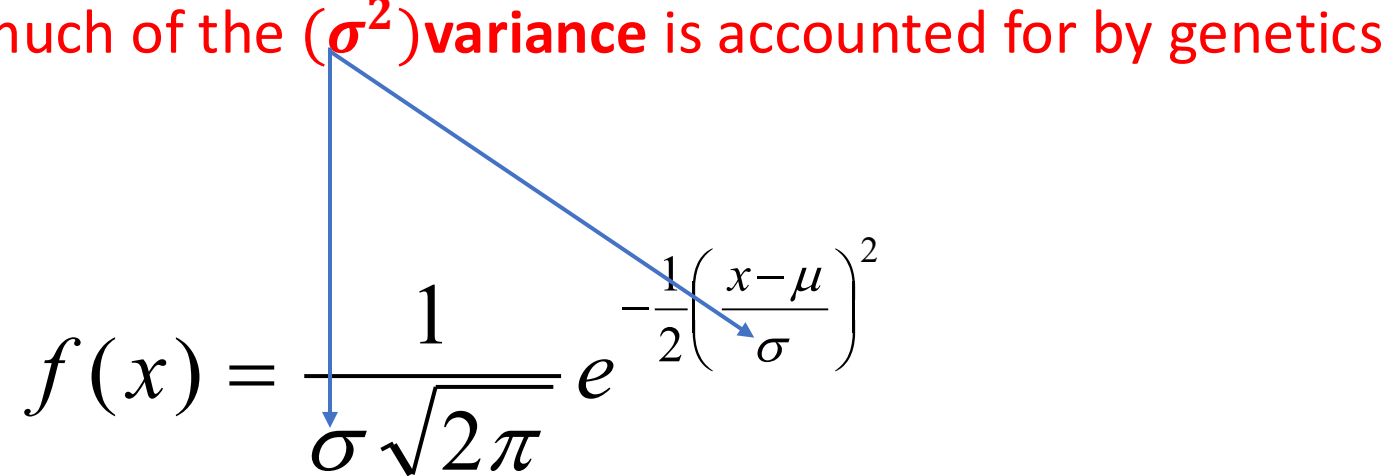
IQ is partially heritable

The size of that component* is the subject of debate, many of the historical studies are flawed

IQ is not entirely heritable

Even the degree of heritability is not constant:
in poor families IQ is less heritable! (Turkheimer)

* We mean how much of the (σ^2) **variance** is accounted for by genetics



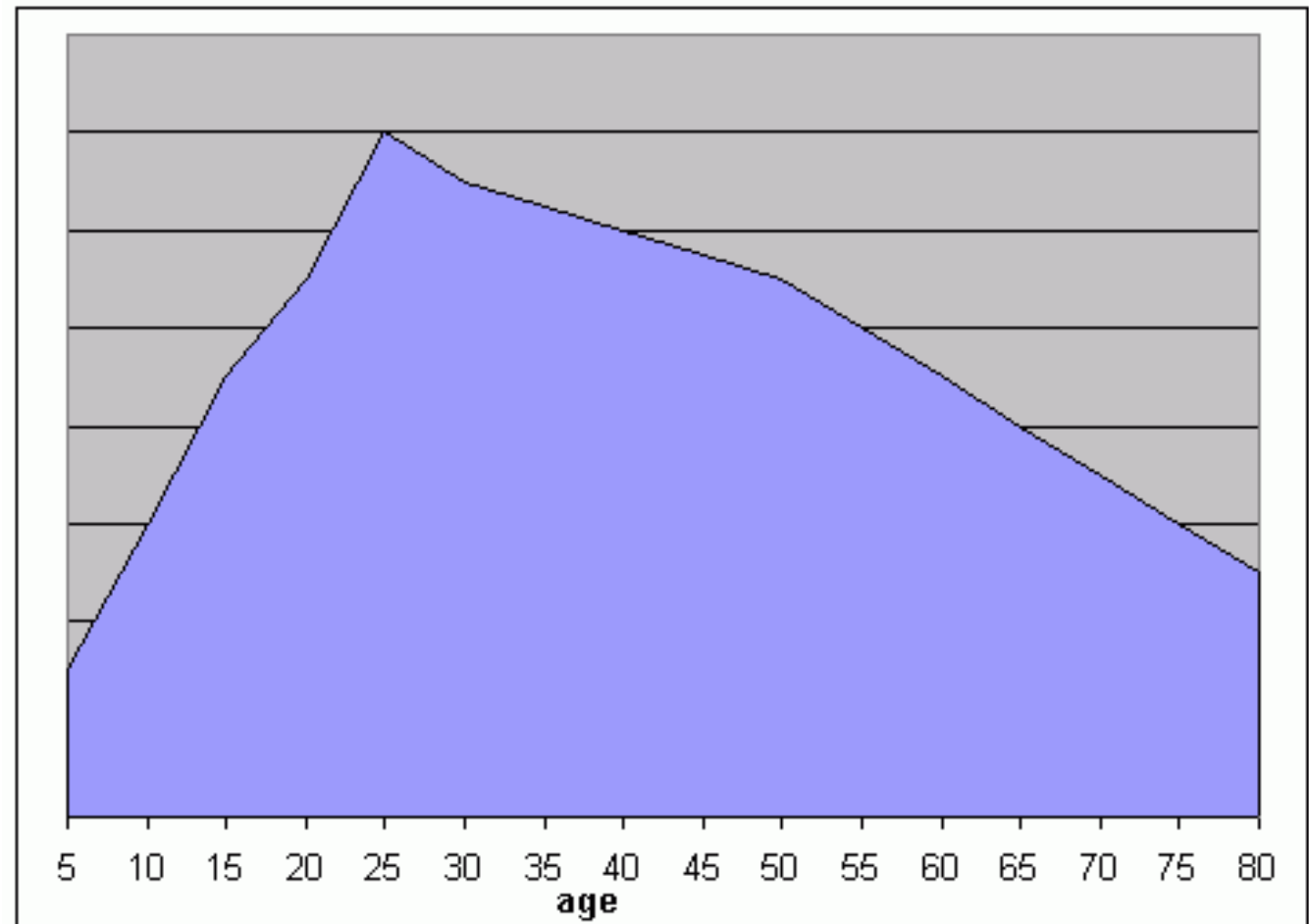
The diagram consists of a blue triangle. A vertical blue line with an arrow at the bottom points from the σ^2 in the text above to the σ in the denominator of the normal distribution formula. A diagonal blue line with an arrow at the end points from the same σ^2 in the text to the σ in the denominator of the exponent's fraction. A horizontal blue line connects the two σ terms.

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2}$$

IQ and age

Some evidence that IQ peaks at **25** and then declines (IQ tests include age as a variable and compensate for this giving a stable score – A.Binet)

The Flynn effect?



IQ and gender

Men have more brain cells! 😊

So do **elephants** 😞

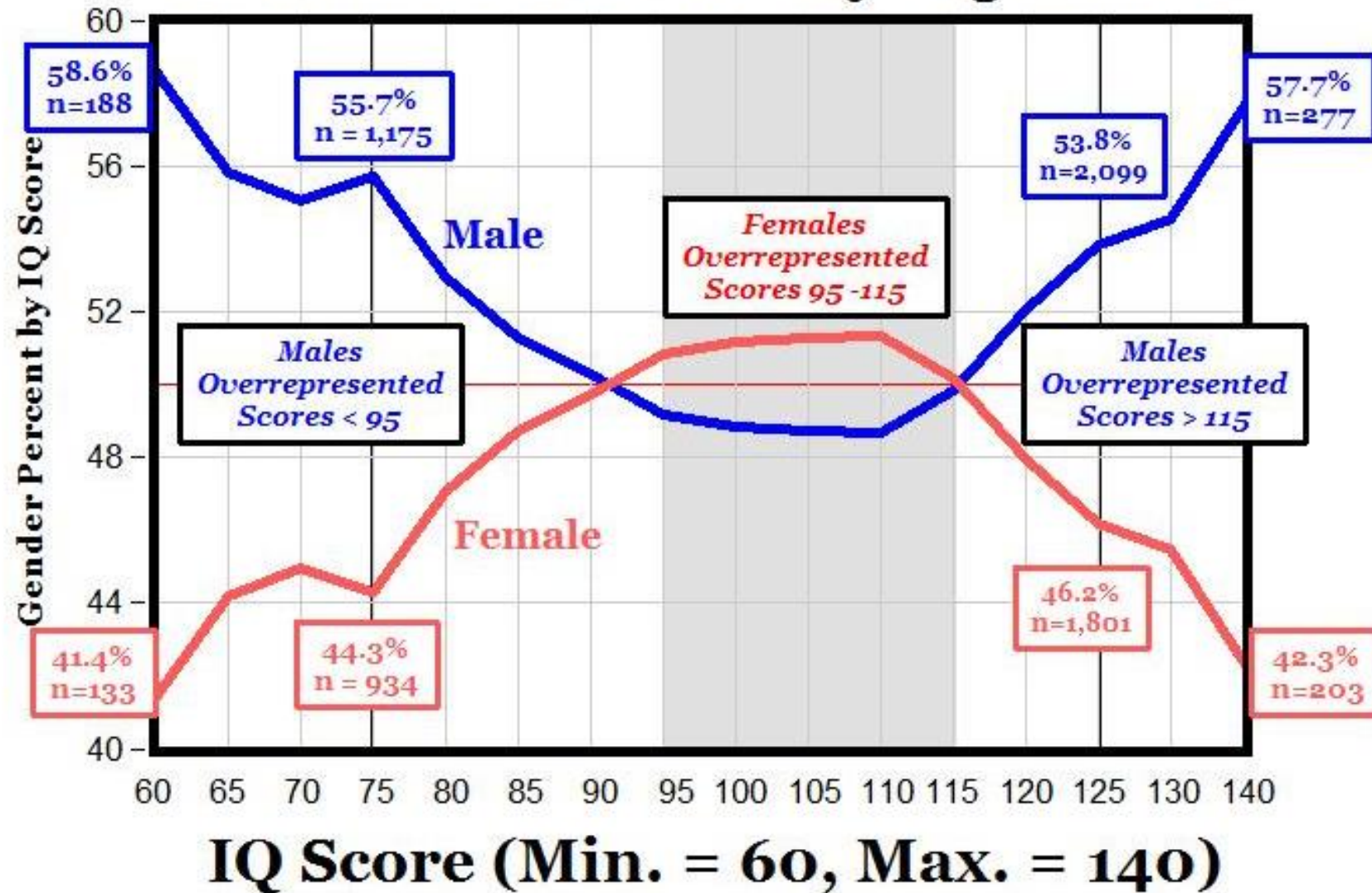
Difference seems to be accounted for by the difference expected to larger body size

Whether men or women have higher mean **IQ** scores varies by country so likely a cultural artefact. Typically gender difference in the tails of the distribution.

IQ Scores in Scotland (1932)

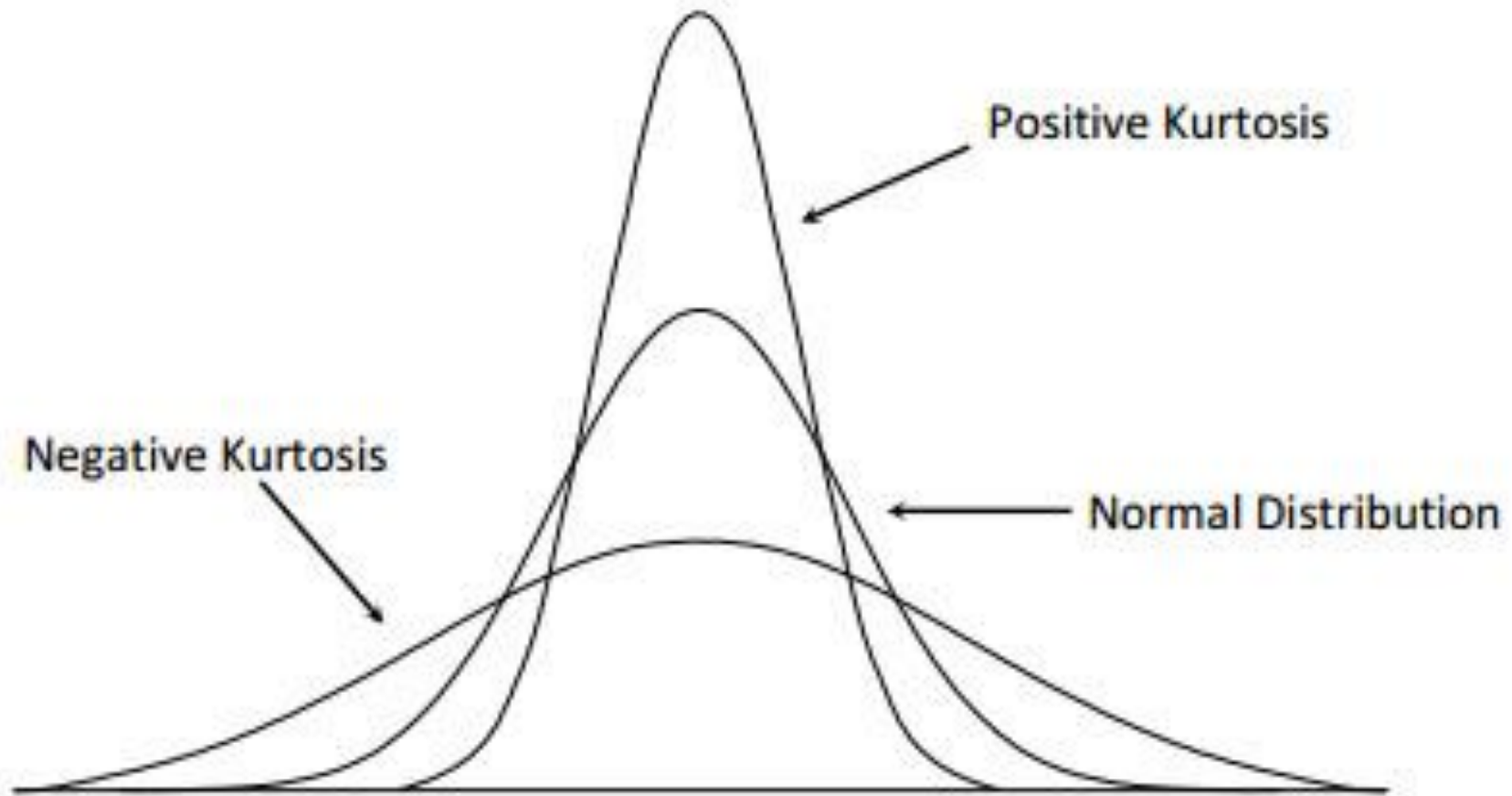
Sample = 79,376 11-year Olds

Gender Percent by IQ Score

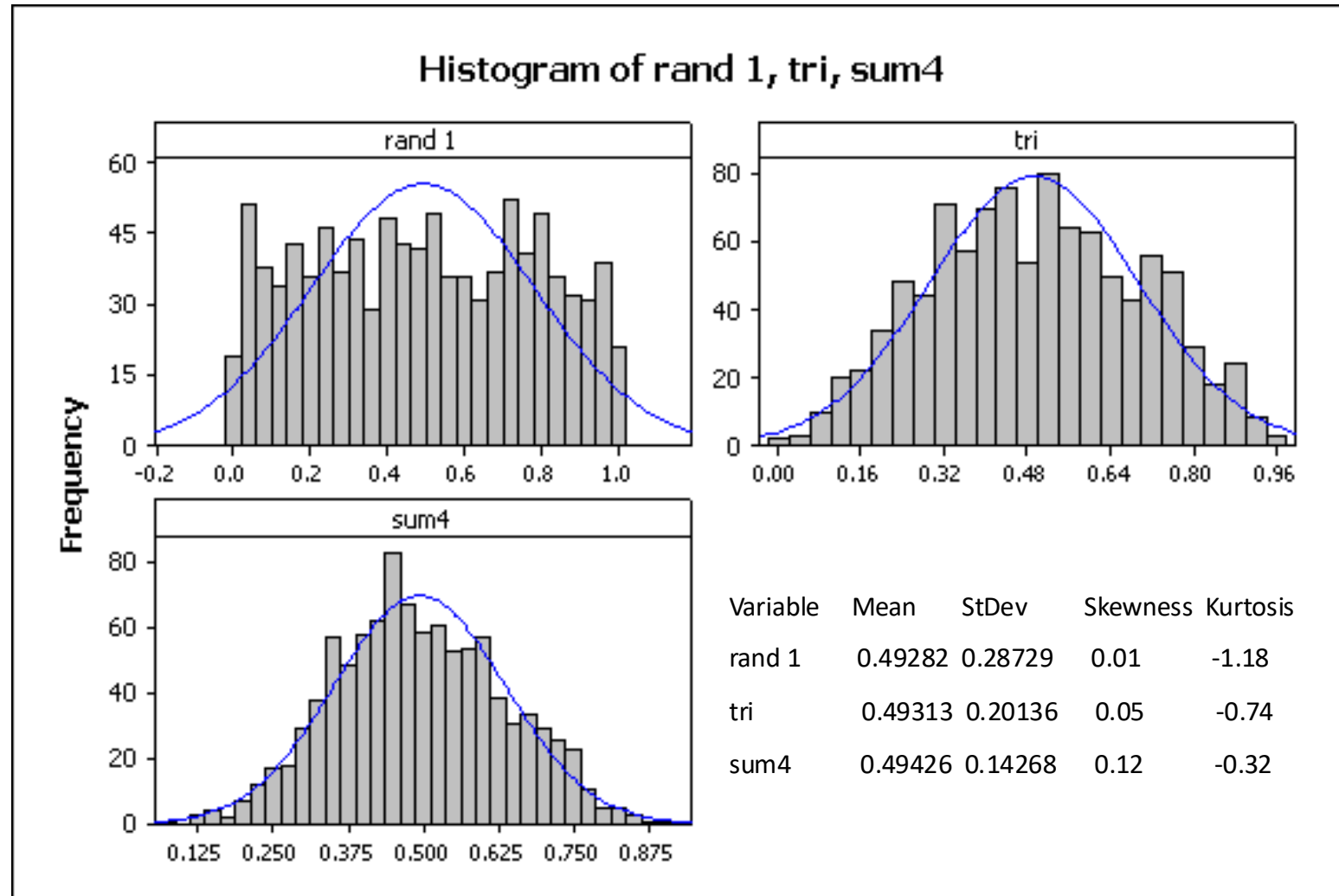


Source: Scottish Mental Survey 1932

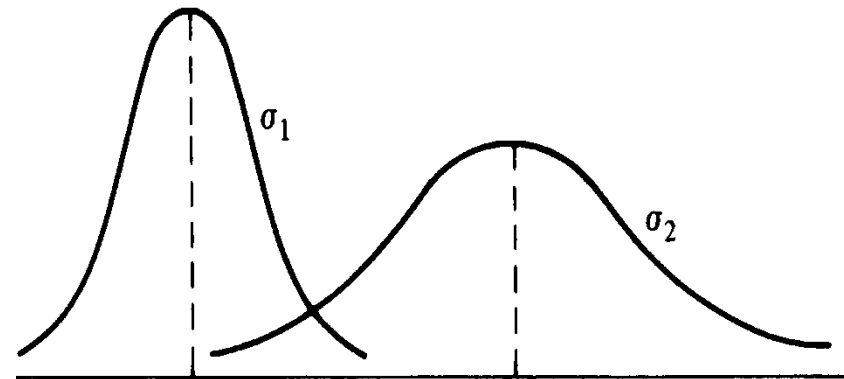
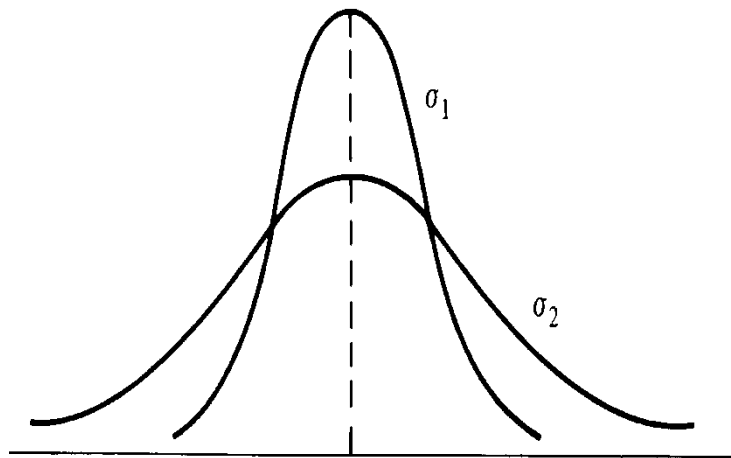
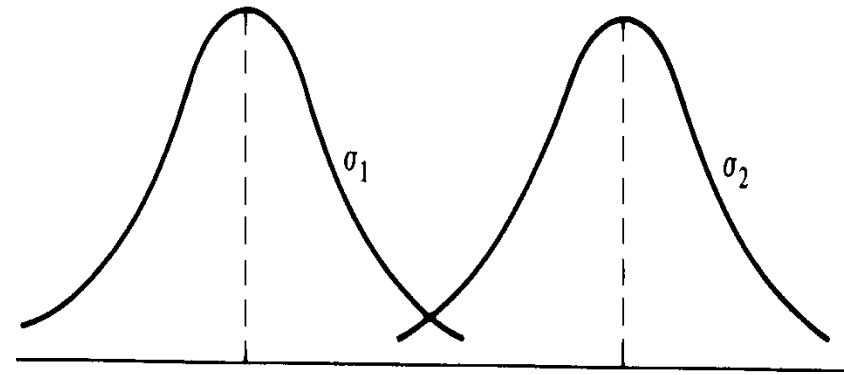
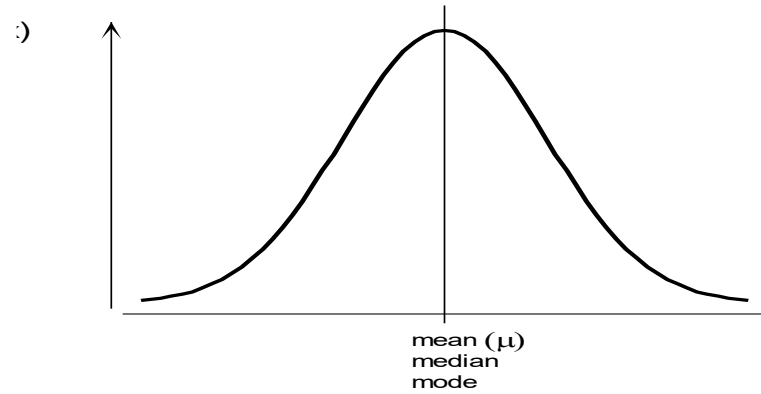
Kurtosis & the normal distribution



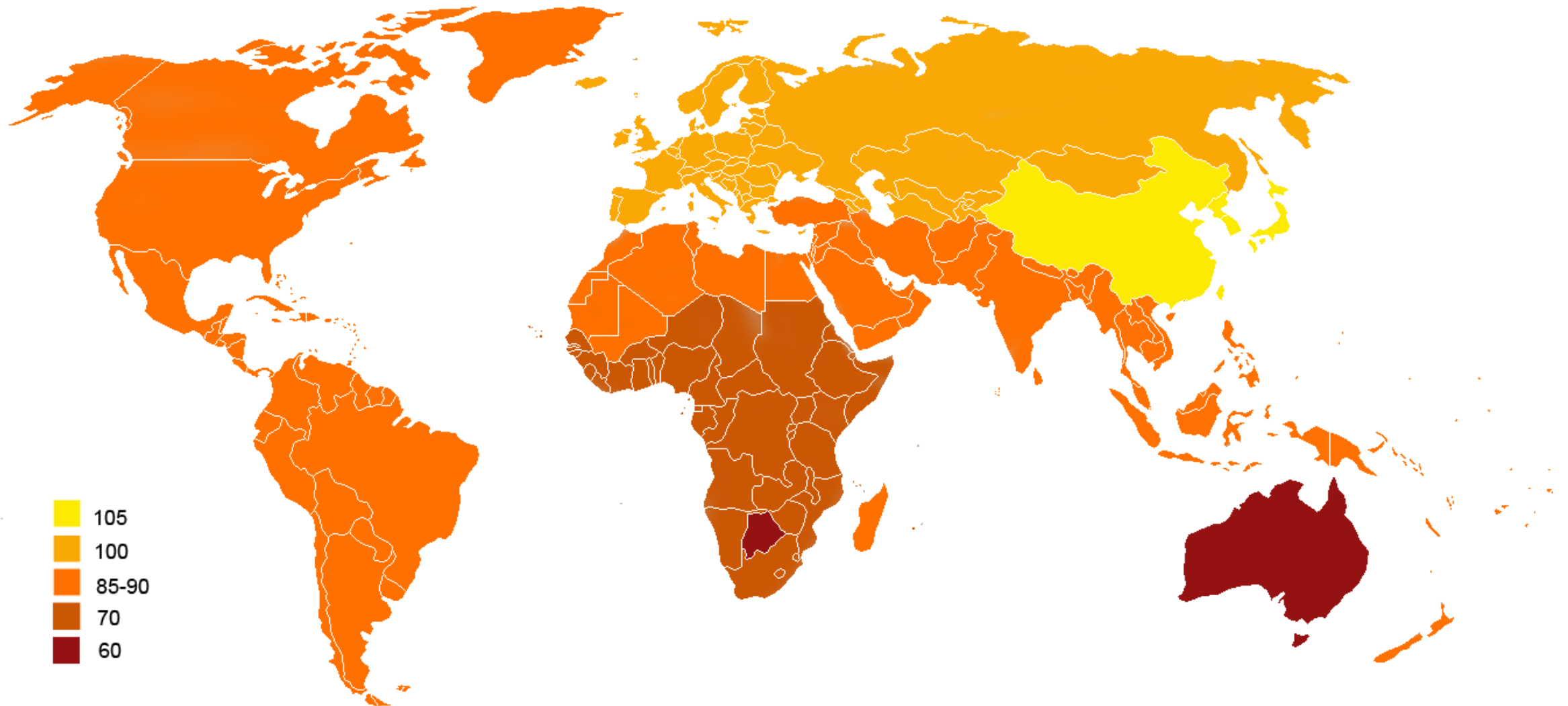
Kurtosis Vs Skewness



The normal distribution and σ



Global IQ map (Lynn 2006)



Sample criticism of map (Wicherts et al 2010)

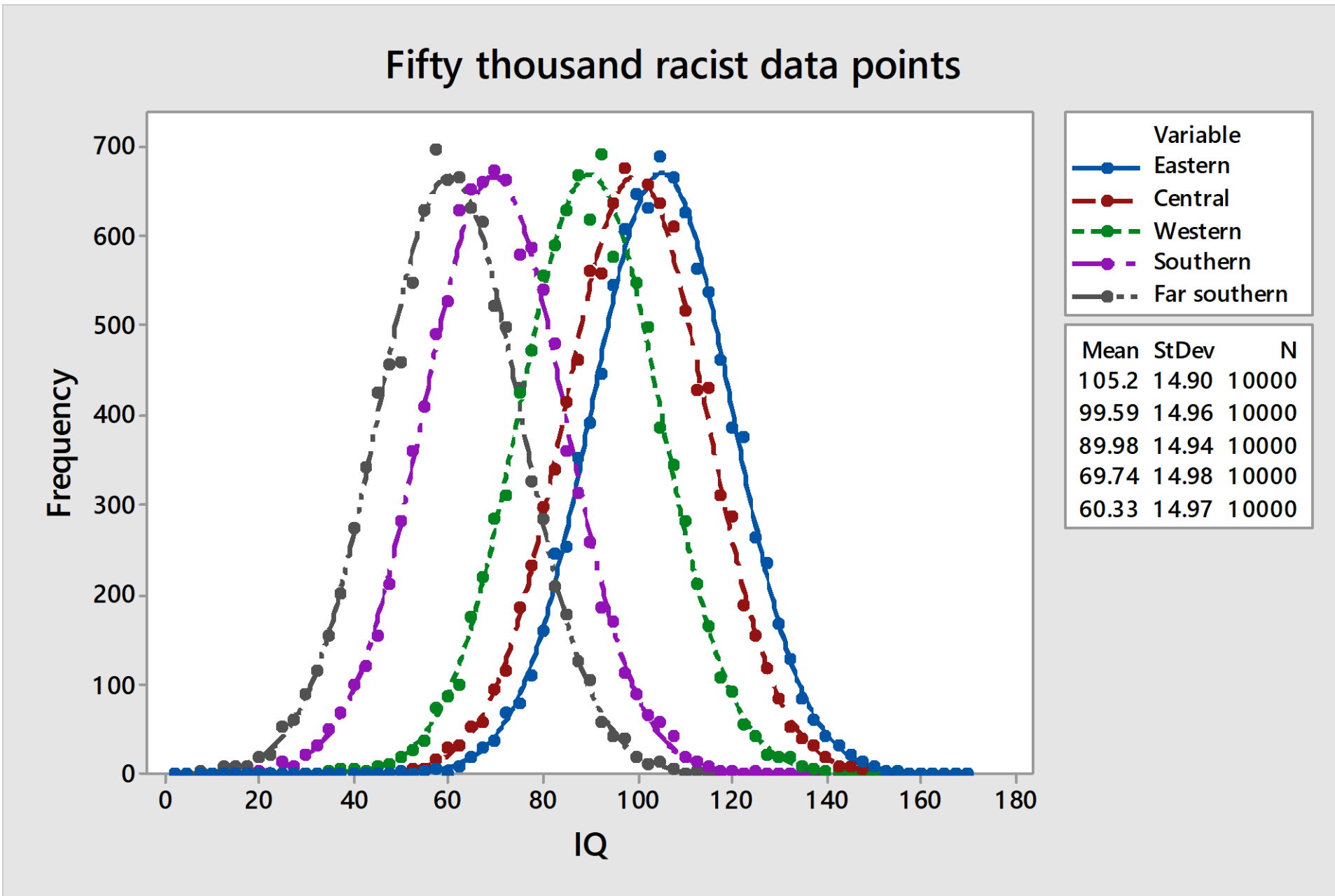
Lynn excluded samples of Africans who average IQs above 75 because they deemed these samples unrepresentative on the basis of the samples' relatively high IQs

The “data point” that Lynn used for the lowest IQ estimate, Equatorial Guinea, was actually the mean IQ of a group of Spanish children in a home for the developmentally disabled in Spain.

The samples, considered by Lynn include... 17 Black South Africans carefully selected for their illiteracy and a sample of uneducated adults from a primitive tribe in the then Belgian Congo in the 1950s.

The SPM test format was rather confusing to the test-takers, and that the test did not meet the standards of valid measurement.

In terms of the graphs we've been using:

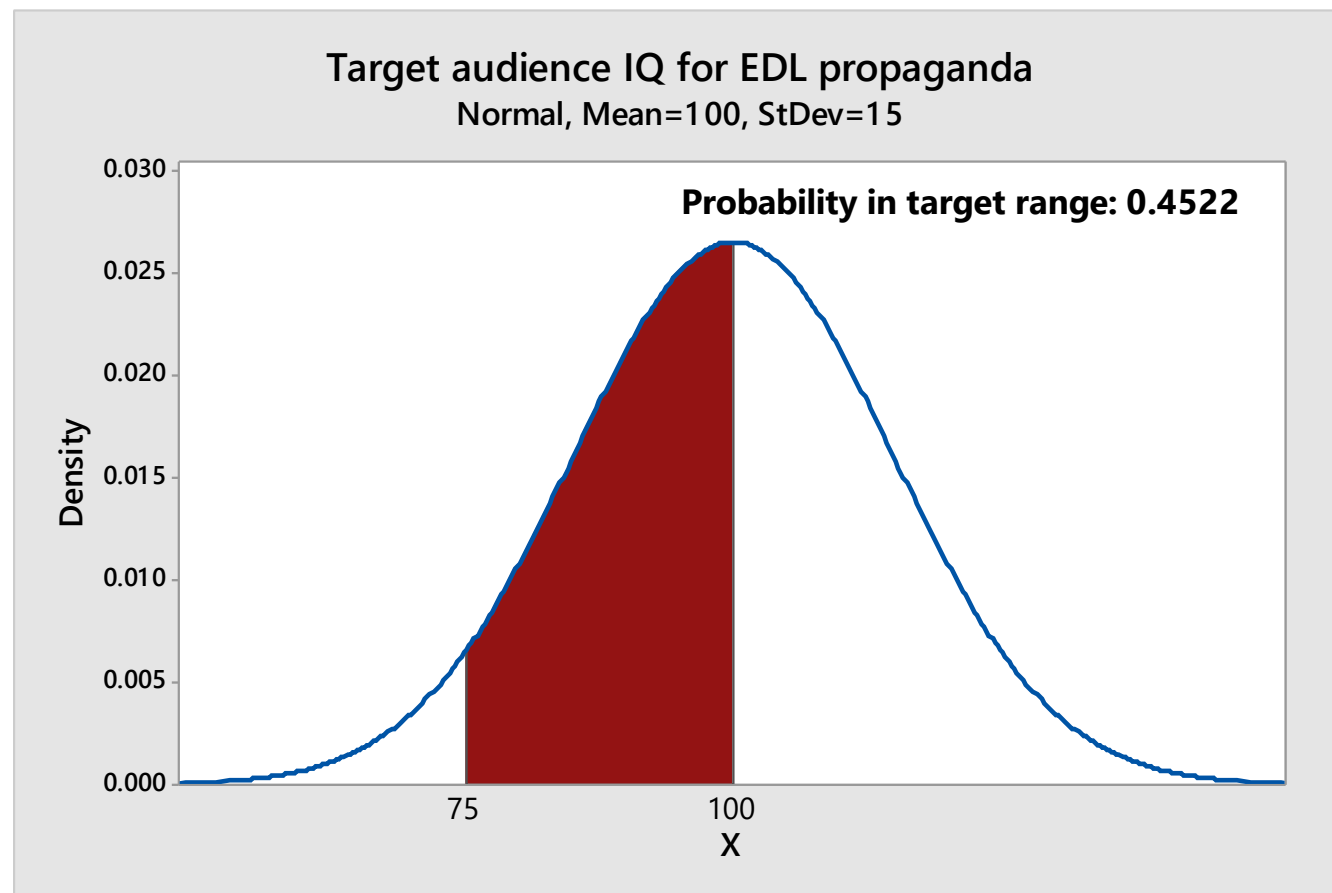


How to use it?

Allows us to calculate probability density using our sample

We use this to make an inference about the population our sample came from

We use the probability density:



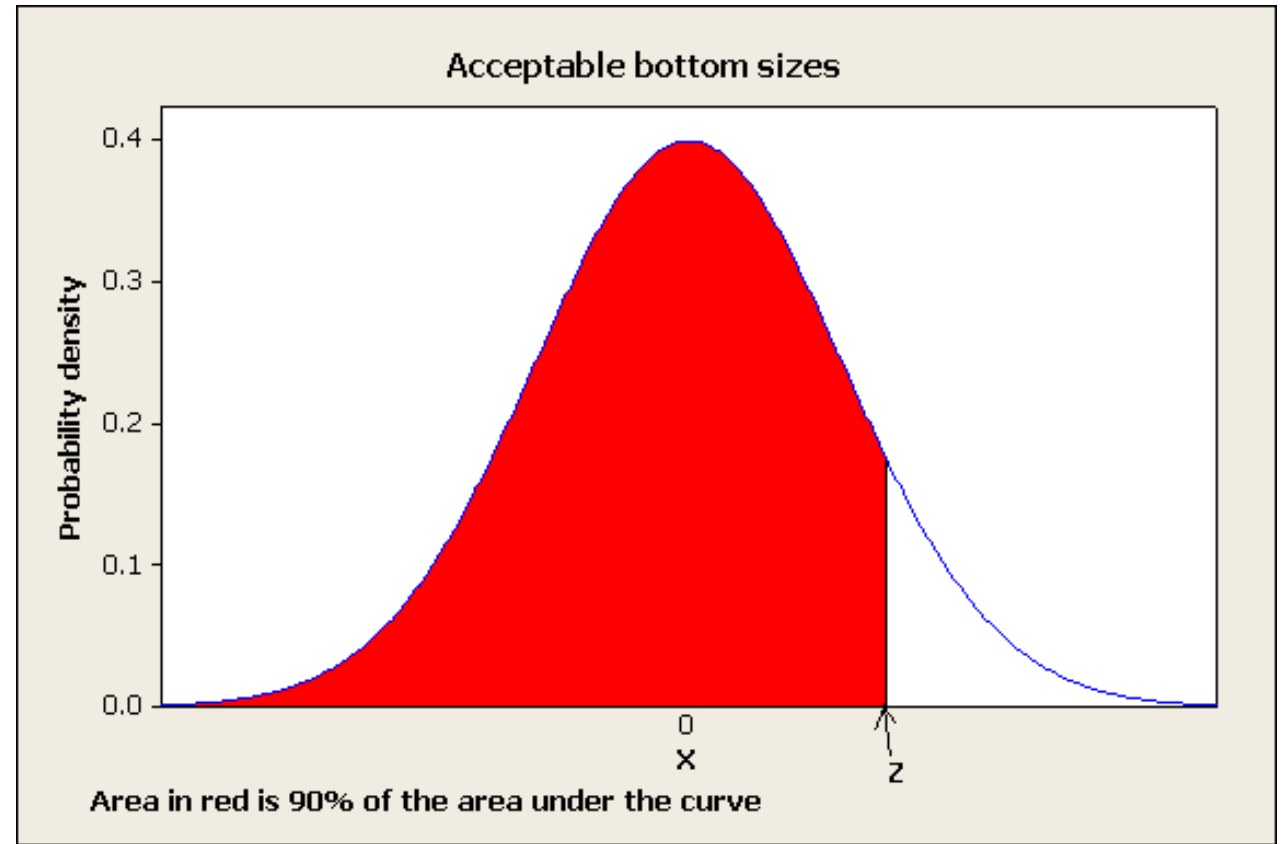
Probability density Example

We're designing a new airplane.

We want to account for people having big bottoms

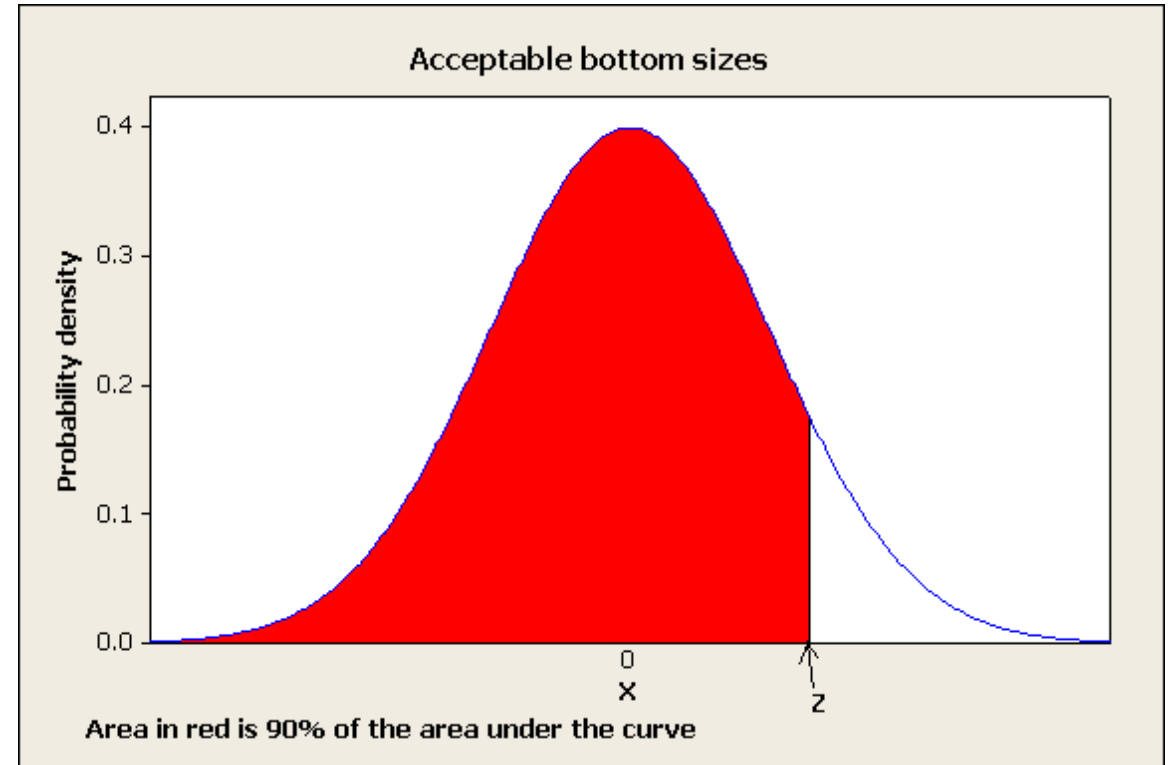
We also want to fit in as many seats as possible

We decide we'll make the seats big enough for 90% of people and make the biggest 10% buy two tickets



To find what number Z is we need to do a study to find the average bottom size and the standard deviation

Please line up now for measurement.



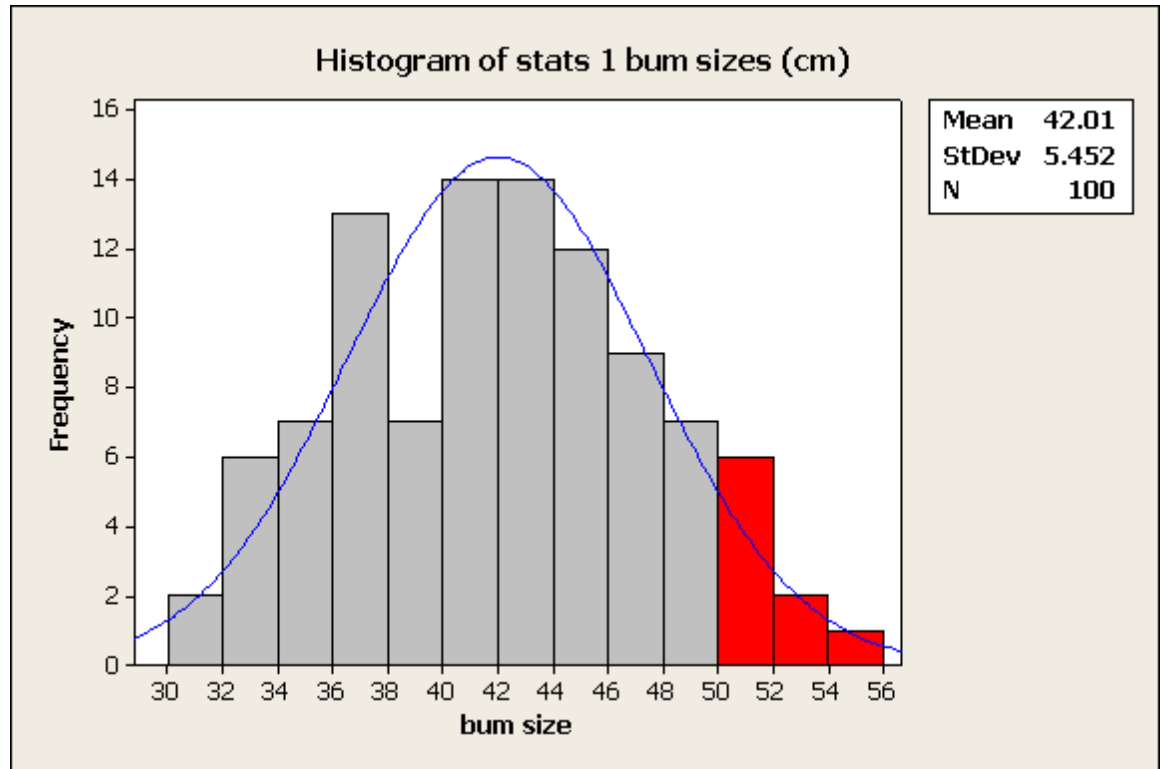
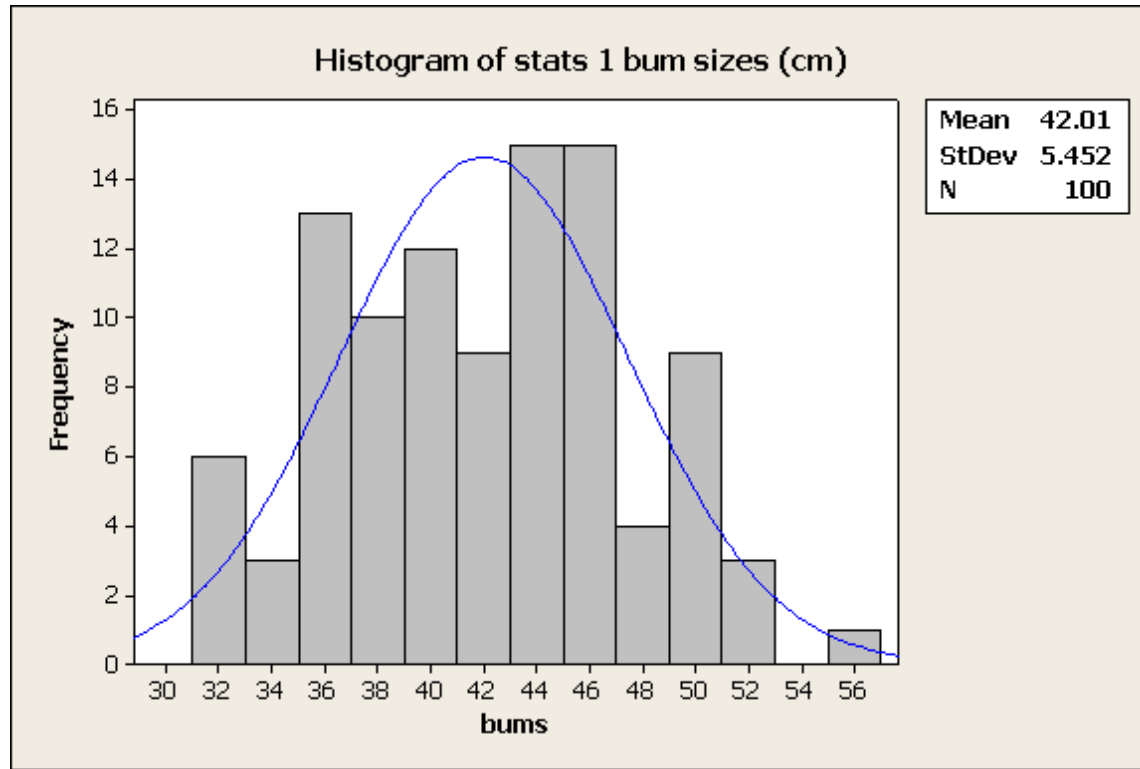
Results

O.K. lockdown has been tough on our diets.

So your average bum width is 42cm with a standard deviation of 6cm

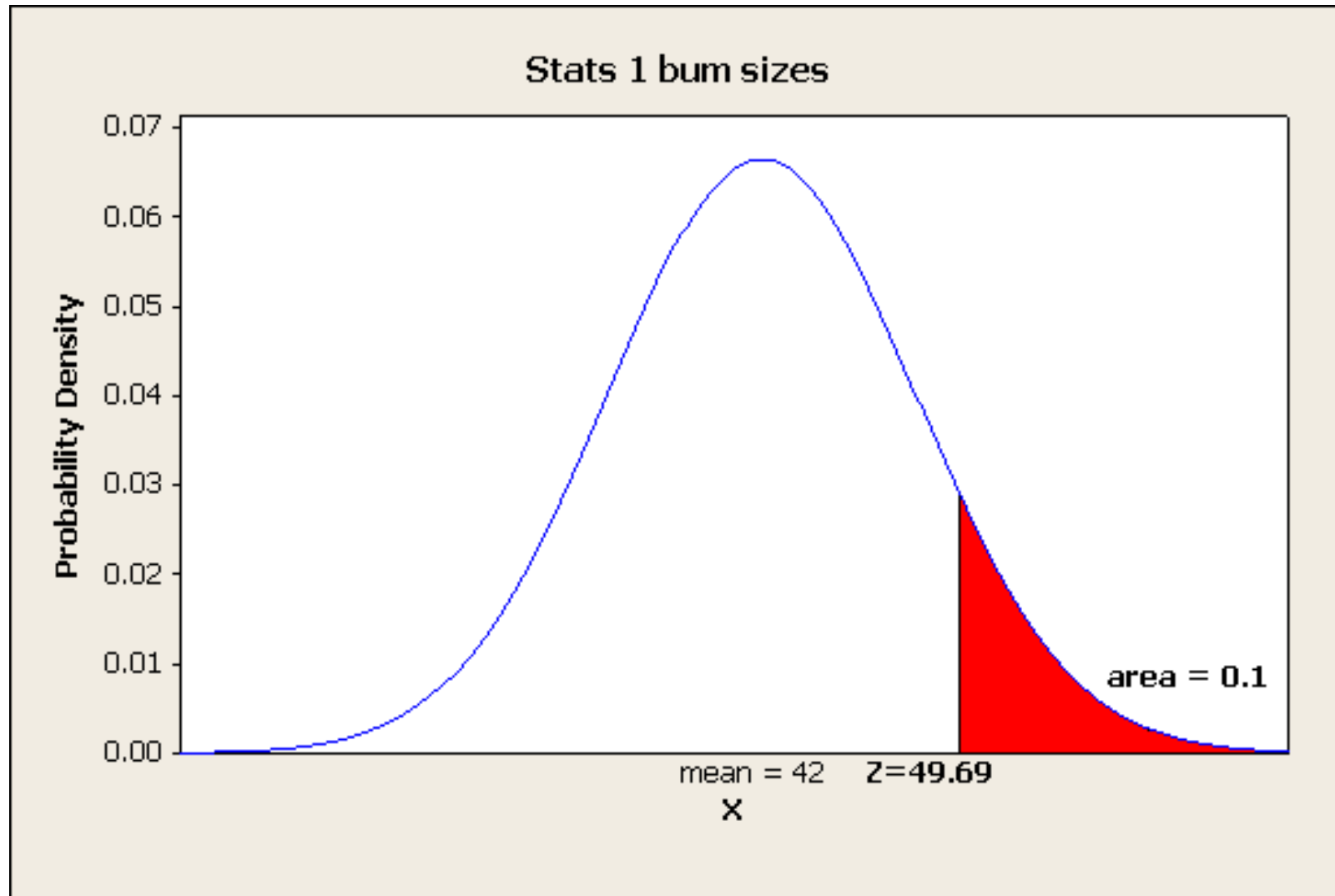
Foundation bum size (cm)				
33.5	37.0	50.5	39.4	46.8
49.8	36.4	45.8	40.2	45.1
37.9	31.6	45.5	44.8	36.8
35.8	38.7	43.3	49.2	36.6
44.2	43.1	40.2	44.5	35.0
50.6	46.2	55.2	52.1	42.2
41.9	42.8	52.6	40.0	32.4
38.1	49.0	36.8	43.9	51.5
51.0	37.1	33.2	44.5	46.3
36.0	46.3	46.5	36.8	40.3
43.7	39.5	48.1	43.3	35.9
46.4	40.2	49.1	37.0	35.7
45.5	44.2	42.0	50.1	41.8
37.7	43.4	45.6	48.5	41.6
43.7	39.5	32.9	41.6	43.5
46.2	32.5	34.7	38.3	36.0
32.4	37.4	43.3	40.2	37.2
40.4	46.4	40.4	50.2	45.2
48.8	45.6	38.5	31.7	35.0
47.1	42.1	41.7	43.4	40.6

Histograms



We can graph this data – note the difference in appearance when we change the bucketing used.

We can ask Minitab to show us the 'real' answer using the standard normal distribution:



Summary

The Normal Distribution is amazing, mathematically beautiful and incredibly useful

You can use it to be racist if you like, but it doesn't make you right

Scientists can be lazy, prejudiced and wrong but still get published – develop your own analytical skills and be critical of what you are told.